QINGLIN MA

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RESEARCH

- The interactions between the large-scale structure and halos/galaxies.
- The formation and evolution of the dark matter halo and its coevolution with galaxy.
- Connecting galaxies to dark matter halos using spectroscopic survey.

EDUCATION

Ph.D. in Astronomy, **Tsinghua University** *Sept. 2021 - July. 2027 (expected)*
Supervisors: Cheng Li. Thesis: Evolution of the dark matter halo and its impact on the galaxy formation

Visiting Scholar, **Leiden Observatory** *Nov. 2025 - July. 2026*
Supervisors: Joop Schaye.

B.S. in Astronomy, **Sun Yat-sen University** *Sept. 2017 - Jun. 2021*
Supervisors: Xiaodong Li.

HONORS

Tsinghua Scholarship for the Visiting Scholar *Nov. 2025 - Mar. 2026*

1st Comprehensive Scholarships 2025

Excellent Oral presentation in the 26th Guo Shoujing Symposium *2024*

Excellent Graduate Student of Sun Yat-sen University *2021*

TEACHING

- Observational Cosmology. Teaching Assistant, Tsinghua University, Autumn 2022
- Machine learning in astronomy Teaching Assistant, Sun Yat-sen University, Spring 2017

SERVICE

- Co-leader of weekly Text club at Department of Astronomy in Tsinghua University *2022-2025*
- Volunteer of Conference "The hundred-billion-year evolution law of galaxy ecosystems" *Jul, 2025*
- Co-organizer of weekly Galaxy-Cosmology Seminar at Department of Astronomy in Tsinghua University *2023*
- Faculty Candidate Interview Committee at Department of Astronomy in Tsinghua University, Phd Representative *2023*

CONFERENCE TALKS

- European Astronomical Society Annual Meeting 2026 **(Contributed talk)** *Lausanne, 2026. Jul*
  Three-stage assembly of massive galaxies in the COLIBRE simulation
- European Astronomical Society Annual Meeting 2026 **(Contributed talk)** *Lausanne, 2026. Jul*
  Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias
- 12th Sino-German Meeting **(Lightning talk)** *Chengdu, 2025. Sep*
  The coevolution between galaxies and dark matter halos in FIRE-2 Simulations
- The hundred-billion-year evolution law of galaxy ecosystems **(Lightning talk)** *Beijing, 2025. Jul*
  The coevolution between galaxies and dark matter halos in FIRE-2 Simulations

- Expanding the boundaries of dark matter halo **(Contributed talk)** *Shanghai, 2025. Mar*
  Dependence of dark matter halo assembly bias on halo definition and orientation
- PFS collaboration meeting **(Lightning talk)** *Tokyo, 2025. Jan*
  Dependence of dark matter halo assembly bias on halo definition and orientation
- 2024 PKU International PhD Student Forum **(Contributed talk)** *Beijing, 2024. Dec*
  Lopsided and Bulging Distribution of Satellites around Paired Central Galaxies
- The 26th Guo Shoujing Symposium **(Contributed talk)** *Nanjing, 2024. Mar*
  Dependence of dark matter halo assembly bias on halo definition and orientation
- 2nd Shanghai Assembly on Cosmology and Structure Formation **(Contributed talk)** *Shanghai, 2023. Oct*
  Dependence of dark matter halo assembly bias on halo definition and orientation

SEMINAR

- Lunch talk, Leiden Observatory **(invited)** *Netherlands, 2026. Jan*
  Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias
- Lightning talk, Department of Astronomy in Tsinghua University *Chengdu, 2025. Sep*
  The coevolution between galaxies and dark matter halos in FIRE-2 Simulations

PUBLICATIONS

- 8 publications; 4 as the first/corresponding author; open in NASA/ADS

First and Corresponding† author papers:

1. **Qinglin Ma**, Yangyao Chen† & Houjun Mo, **MNRAS** 548 3 (2026) [ArXiv: 2509.16138]
   Two-phase formation of galaxies: the coevolution between galaxies and dark matter haloes
2. **Qinglin Ma**, Cheng Li†, Yangyao Chen & Houjun Mo, submitted to **MNRAS** (2025) [ArXiv: 2512.17742]
   Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias
3. **Qinglin Ma**, Cheng Li† & Yanhan Guo, **APJ** 922 1 (2025) [ArXiv: 2502.14668]
   Lopsided and Bulging Distribution of Satellites around Paired Halos. II. 3D Analysis and Dependence on Projection and Selection Effects
4. **Qinglin Ma**, Guo, Yiqing, Li, Xiao-Dong†, Wang, Xin, Miao, Haitao, Li, Zhigang, Sabiu, Cristiano G & Park, Hyunbae, **APJ** 890 2 (2020) [ArXiv: 1908.10595]
   Cosmological Constraints from the Redshift Dependence of the Alcock-Paczynski Effect: Possibility of Estimating the Nonlinear Systematics Using Fast Simulations

Co-author papers:

1. Ce Gao, Cheng Li, Houjun Mo, Jiacheng Meng, **Qinglin Ma**, Xiaohu Yang, Yizhou Gu, Qingyang Li, submitted to **MNRAS** (2026) [ArXiv: 2604.04794]
   Galaxy Populations in Groups and Clusters: II. Conditional Luminosity Functions at Redshifts from $z \sim 1$ to $z \sim 0$
2. Yanhan Guo, **Qinglin Ma** & Cheng Li, **APJ** 922 1 (2025) [ArXiv: 2502.14666]
   Lopsided and Bulging Distribution of Satellites Around Paired Halos. I. Observational Measurements and Comparison with Halo-based Models
3. Shuren Zhou, Zhenjie Liu, **Qinglin Ma**, Yu Liu, Le Zhang, Xiao-Dong Li, Yang Wang, Xin Wang, Yu Yu, Hao-Ran Yu & Yi Zheng, **MNRAS** 512 3 (2021) [ArXiv: 2108.12568]
   Sensitivity tests of cosmic velocity fields to massive neutrinos
4. Yizhao Yang, Haitao Miao, **Qinglin Ma**, Miaoxin Liu, Cristiano G Sabiu., Jaime Forero-Romero, Yuanzhu Huang, Limin Lai, Qiyue Qian, Qiyue Zheng & Xiao-Dong Li, **APJ** 900 1 (2021) [ArXiv: 2007.03150]
   Using the Mark Weighted Correlation Functions to Improve the Constraints on Cosmological Parameters

REFERENCES

Prof. Cheng Li *Tsinghua university*
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Prof. Houjun Mo *University of Massachusetts, Amherst*
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Prof. Joop Schaye *Leiden Observatory*
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